



CLINICAL PROGRAMMING BOOTCAMP • MODULE 12

Define-XML & Submission

What actually goes to the FDA — and in what format

Your datasets are only part of the package. This module covers the metadata that describes them, and the file format the regulator actually accepts.

Concept module — closes the submission story.

Datasets are one of five things

SDTM datasets

the tabulation data you built — as XPT v5 files

`dm.xpt · ae.xpt · vs.xpt · ...`

ADaM datasets

analysis-ready data, derived FROM SDTM (a later course)

`adsl.xpt · adae.xpt`

define.xml

the machine-readable metadata describing every dataset and variable

`define.xml`

Annotated CRF

the blank CRF marked up with the SDTM variable each field maps to

`acrf.pdf`

Reviewer's Guides

cSDRG and ADRG — the human explanation of the data

`csdrg.pdf`

You built the first box. This module is about the middle three — the metadata and format that make the datasets usable by someone who has never seen your study.

SAS Transport v5 — and the constraints it forces

The FDA does not accept .sas7bdat. It accepts XPORT version 5 (XPT) — an open, decades-old format any system can read. That longevity is the point: a submission must be readable in 25 years.

SAS · writing an XPT v5 file		
<pre>libname xptout xport "&outpath/dm.xpt"; /* the XPORT engine */ data xptout.dm; set sdtm.dm; run; libname xptout clear;</pre>		
Dataset & variable names	≤ 8 characters, uppercase	why AESEQ, LBTESTCD, not a long name
Variable labels	≤ 40 characters	the short QLABEL you have been writing
Character values	≤ 200 bytes	long text must be split across --TESTCD groups
Numeric dates	not allowed — dates are CHARACTER ISO 8601	why every --DTC is a string, never a SAS date

Every one of these explains a design choice you already met. XPT v5 is WHY SDTM looks the way it does.

define.xml — the data's instruction manual

Without it, your datasets are a pile of columns with cryptic names.

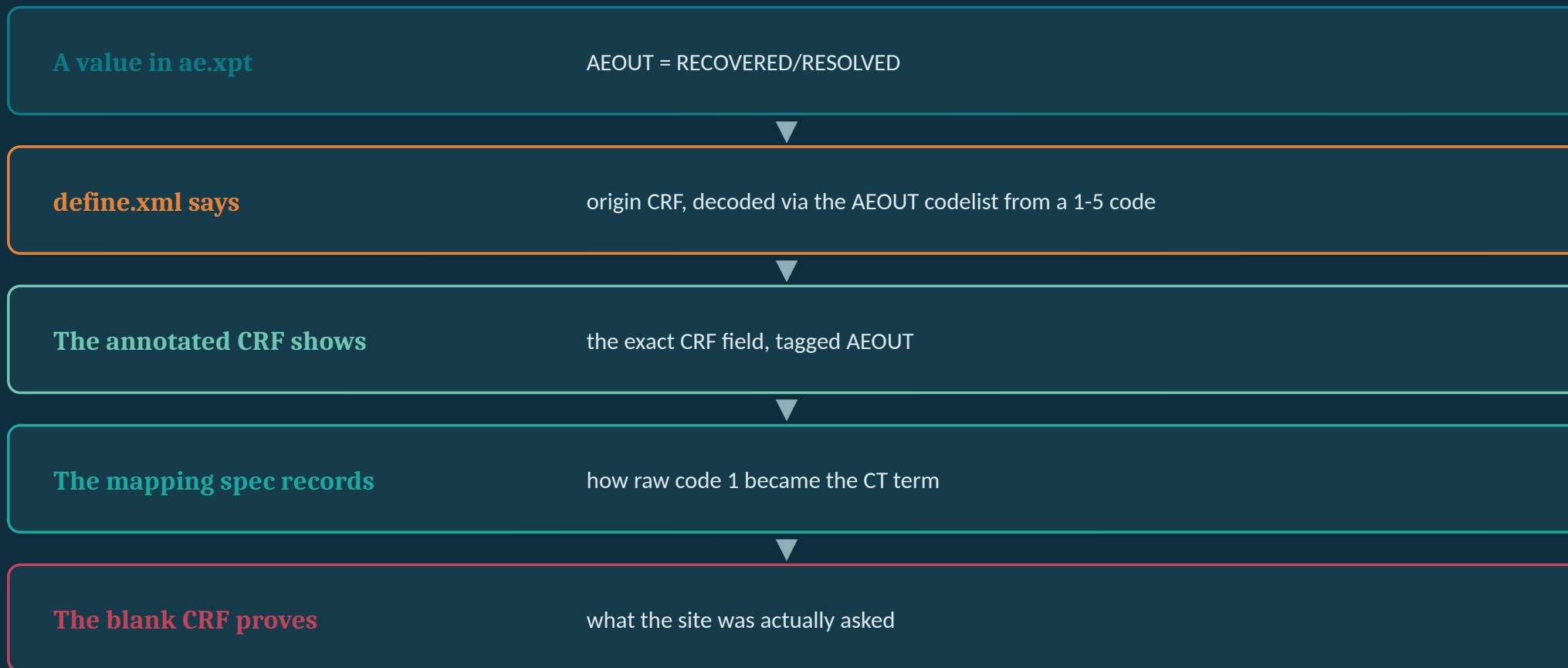
define.xml is machine-readable metadata that describes EVERY dataset and EVERY variable: its label, type, length, the codelist it draws from, where it came from, and how it was derived. A reviewer opens define.xml FIRST, and their tools read it to make sense of everything else.

For every variable, define.xml records:

Metadata	For AEOUT, say	Why the reviewer needs it
Label & type	"Outcome of AE", text	to read the column at all
Codelist	→ the AEOUT codelist	to know the allowed values
Origin	CRF / DERIVED / ASSIGNED	to trust where it came from
Derivation	"decoded from raw code 1-5"	to reproduce or audit it

Origin is the one people underestimate. It tells the reviewer whether a value was collected on the CRF, derived by a program, or assigned by the sponsor. AETRTEM is DERIVED; AETERM is CRF. That provenance is what makes the data auditable.

From the reviewer's question back to the CRF



A reviewer can start at any submitted value and trace it all the way back to the question the site answered. That chain — value → metadata → aCRF → spec → CRF — is what makes clinical data TRUSTWORTHY, not just tidy.

What to carry out of Day 9

Datasets alone are not a submission

define.xml, the aCRF and the reviewer's guides are not optional extras — they are what make the data usable.

The format shapes the standard

8-char names, character dates, short labels — XPT v5 is why SDTM looks the way it does.

Origin is provenance

every value is CRF, DERIVED or ASSIGNED, and define.xml records which. That is what makes it auditable.

Everything must trace back

value → define.xml → aCRF → spec → CRF. If a value cannot be traced to a question, it cannot be trusted.